

LAB	PRACTICALS
23	Implementation of JOIN in SQL Create the following tables and perform the following queries: Part – A --1. Combine information from student and result table using cross join or Cartesian product. <code>SELECT * FROM Student CROSS JOIN Result;</code> --2. Display Rno, Name, Branch and SPI of CE branch's student only. <code>SELECT S.RNO, S.NAME, S.BRANCH, R.SPI FROM STUDENT S INNER JOIN RESULT R ON S.RNO = R.RNO WHERE S.BRANCH = 'CE';</code> --3. Display Rno, Name, Branch and SPI of other than EC branch's student only. <code>SELECT S.RNO, S.NAME, S.BRANCH, R.SPI FROM STUDENT S JOIN RESULT R ON S.RNO = R.RNO WHERE S.BRANCH != 'EC';</code> --4. Display average result of each branch. <code>SELECT S.BRANCH, AVG(R.SPI) AS AVG FROM STUDENT S JOIN RESULT R ON S.RNO = R.RNO GROUP BY S.BRANCH;</code> --5. Display average result of each branch and sort them in ascending order by SPI. <code>SELECT S.BRANCH, AVG(R.SPI) AS AVG FROM STUDENT S JOIN RESULT R ON S.RNO = R.RNO GROUP BY S.BRANCH ORDER BY AVG(R.SPI);</code> Part – B --6. Display average result of CE and ME branch. <code>SELECT S.BRANCH, AVG(R.SPI) AS AVG FROM STUDENT S JOIN RESULT R ON S.RNO = R.RNO WHERE S.BRANCH IN ('CE', 'ME') GROUP BY S.BRANCH;</code>

--7. Perform the left outer join on Student and Result tables.

```
SELECT * FROM  
STUDENT LEFT OUTER JOIN RESULT  
ON STUDENT.RNO = RESULT.RNO;
```

--8. Perform the right outer join on Student and Result tables.

```
SELECT * FROM  
STUDENT RIGHT OUTER JOIN RESULT  
ON STUDENT.RNO = RESULT.RNO;
```

--9. Perform the full outer join on Student and Result tables.

```
SELECT * FROM  
STUDENT FULL OUTER JOIN RESULT  
ON STUDENT.RNO = RESULT.RNO;
```

--10. Retrieve the names of employee along with their manager's name from the Employee table.

```
SELECT  
    E1.EMPLOYEENO,  
    E1.NAME AS EMPLOYEEName,  
    E2.NAME AS MANAGERName  
FROM EMPLOYEE E1 LEFT JOIN EMPLOYEE E2  
ON E1.MANAGERNO = E2.EMPLOYEENO;
```

Part – C

--11. Display maximum result of each branch.

```
SELECT S.BRANCH, MAX(R.SPI) AS MAX_SPI  
FROM STUDENT S JOIN RESULT R  
ON S.RNO = R.RNO  
GROUP BY S.BRANCH;
```

--12. Display minimum result of each branch.

```
SELECT S.BRANCH, MIN(R.SPI) AS MIN_SPI  
FROM STUDENT S JOIN RESULT R  
ON S.RNO = R.RNO  
GROUP BY S.BRANCH;
```

--13. Display name of student whose result is not generated.

```
SELECT S.NAME  
FROM STUDENT S LEFT OUTER JOIN RESULT R  
ON S.RNO = R.RNO  
WHERE R.RNO IS NULL;
```

	--14. Display overall average result. <pre>SELECT AVG(SPI) AS OVERALL_AVERAGE_SPI FROM RESULT;</pre> --15. Display Rno, Name, Branch and SPI of those students whose SPI greater than. <pre>SELECT S.RNO, S.NAME, S.BRANCH, R.SPI FROM STUDENT S INNER JOIN RESULT R ON S.RNO = R.RNO WHERE R.SPI > 8.0;</pre>
24	Implementation of advanced JOIN operations in SQL Create the following tables and perform the following queries: Part – A --1. Display all the villages of Rajkot city. <pre>SELECT V.VILLAGENAME FROM VILLAGE V INNER JOIN CITY C ON V.CITYID = C.CITYID WHERE C.CITYNAME = 'RAJKOT';</pre> --2. Display city along with their villages & pin code. <pre>SELECT C.CITYNAME, V.VILLAGENAME, C.PINCODE FROM VILLAGE V INNER JOIN CITY C ON V.CITYID = C.CITYID;</pre> --3. Display the city having more than one village. <pre>SELECT C.CITYNAME, COUNT(V.VILLAGEID) AS VillageCount FROM CITY C INNER JOIN VILLAGE V ON C.CITYID = V.CITYID GROUP BY C.CITYNAME HAVING COUNT(V.VILLAGEID) > 1;</pre> --4. Display the city having no village. <pre>SELECT C.CITYNAME FROM CITY C LEFT OUTER JOIN VILLAGE V ON C.CITYID = V.CITYID WHERE V.CITYID IS NULL;</pre> --5. Count the total number of villages in each city. <pre>SELECT C.CITYNAME, COUNT(V.VILLAGEID) AS VILLAGECOUNT FROM VILLAGE V JOIN CITY C ON V.CITYID = C.CITYID GROUP BY C.CITYNAME;</pre>

Part – B

--6. Display number of villages of Rajkot city.

```
SELECT C.CITYNAME, COUNT(V.VILLAGEID) AS COUNT
FROM VILLAGE V JOIN CITY C
ON V.CITYID = C.CITYID
WHERE C.CITYNAME='RAJKOT'
GROUP BY C.CITYNAME;
```

--7. Display Rajkot city along with their villages & pin code.

```
SELECT C.CITYNAME, V.VILLAGENAME, C.PINCODE
FROM VILLAGE V JOIN CITY C
ON V.CITYID = C.CITYID
WHERE C.CITYNAME='RAJKOT';
```

--8. Display villages of Rajkot & Junagadh city.

```
SELECT V.VILLAGENAME, C.CITYNAME
FROM VILLAGE V JOIN CITY C
ON V.CITYID = C.CITYID
WHERE C.CITYNAME IN ('RAJKOT', 'JUNAGADH');
```

Part – C

--9. Display number of villages of Surat city.

```
SELECT COUNT(V.VILLAGEID)
FROM VILLAGE V JOIN CITY C
ON V.CITYID = C.CITYID
WHERE C.CITYNAME = 'SURAT';
```

--10. Display city along with their villages & pin code whose Remarks is Good.

```
SELECT C.CITYNAME, V.VILLAGENAME, C.PINCODE
FROM CITY C JOIN VILLAGE V
ON C.CITYID = V.CITYID
WHERE C.REMARKS = 'GOOD';
```

--11. Display the city name of Falla village.

```
SELECT C.CITYNAME
FROM CITY C JOIN VILLAGE V
ON C.CITYID = V.CITYID
WHERE V.VILLAGENAME = 'FALLA';
```